

**ADMINISTRATIVE INFORMATION****FMI No: 25364****FMI Title: SmarTube Phase 2 SW Update for Linux/Pegasus systems**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☒ Mandatory Safety	GEMS - AM:	Issue Date:
☐ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

- ☐ **Staged FMI** **Customer List** provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
- ☐ **Parts** and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
- ☐ **Parts Ordering:** GEMS - Americas, Contact your Region logistics coordinator
GEMS - Europe, Order FMI kits from GPO/EPL Buc
GEMS - Asia, Contact your region logistics coordinator
- ☐ **No Parts** are required to complete this FMI.
- ☐ **Batch with** _____
- To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
- ☐ **Non-Staged** **No Customer List is provided.** Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

This FMI is for CT Linux/Pegasus (GOC2) systems only. Do not install in CT/PET.

Software prerequisite:

- **LightSpeed 3X M4** 308L.8_H3.3M4 or 308L.5_H3.3M4
- **LightSpeed 4X M4** 405L.8_H4.2M4 or 405L.5_H4.2M4
- **HiSpeed Qxi** 103L.5_HSQXIM4 (FMI 25903)
Or 103L.4_HSQXIM4

Unit Tracked:**FMI Instructions Part No.: 5162173-100 Rev 1****Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas

GEMS - Europe

G.E. Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe
FMI Admin. 322
BP34
F78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25364

FMI Title: SmarTube Phase 2 SW Update for Linux/Pegasus systems

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer
Comments

Effectivity

CT systems using Linux/Pegasus. Console numbers to be tracked:

- 2337535 and 2337535-2
- 2366438 (H16 Linux)

Purpose

The purpose of this FMI is to load patch C.13.3 on systems running Linux/Pegasus M4 software. This patch includes all the fixes and enhancements of all prior patches including patch B.2 (optional FMI 25353). When installing patch C.13.3, there is no need to install any other previous patches.

Also the software includes SmarTube Phase 2 features.

Prerequisite

The software should be:

- LightSpeed 3X M4 308L.8_H3.3M4 or 308L.5_H3.3M4
- LightSpeed 4X M4 405L.8_H4.2M4 or 405L.5_H4.2M4
- HiSpeed Qxi 103L.5_HSQXIM4 (FMI 25903)

Or 103L.4_HSQXIM4

- To check the software revision, open unix shell and type command, **check_config**
- This FMI supercedes FMI 25353 for Linux/Pegasus platform.

Note:

Due to enhancements to SmartStep, systems with SmartStep Option must also have the Recon Plus Option installed to avoid potential problems.

To check software options:

Open Unix Shell

Type **swokinstall -p** <Enter>

If the Recon Plus Option is not installed, contact OLC for help in obtaining the option key.

Furnished Materials

Below are the items included in kit P/N 5136943

Qty	Description	Part #
1	FMI 25364 Document	5162173-100
1	Linux/Pegasus M5 software patch C	5154649
1	Linux/Pegasus Tips and Workarounds	5130011-100
1	Customer Letter	5142333-147

Tool Required

Qty	Description	SW Version	Part #
1	FMI 25364 Document		

Personnel Requirements

Travel: 1 Hour Labor 1 Hour Field Engineer: 1

IMPROVEMENTS

The B.2 patch fixes shown below are included in the new C.13.3 patch.

Product Safety Fixes

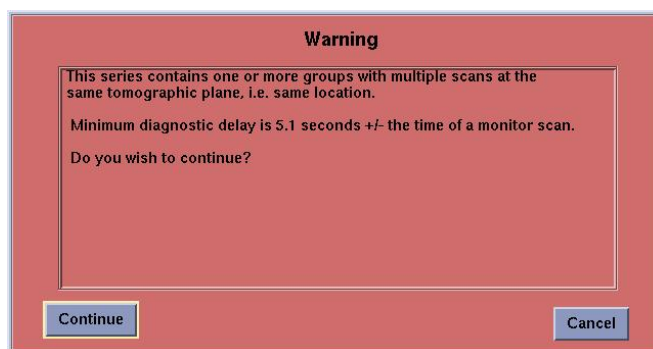
- Scout Image Fails to recon
This was caused by characters added at the end of the Exam Description when the shift or control key was pressed along with the enter key.
- Additional slices added when scans are prescribed with gantry tilt
Improvements have been made in Graphic Rx and ScanRx
- Show Localizer lines are in wrong location if user skips series quickly
- Smart Step Images out of sequence
- Auto mA changes automatically or reduces to 10 mA
This was due to how Noise index was updated when the user changed slice thickness. Prior to this update, the Noise index adjustments were made from the Reference Noise Index. The change is to adjust Noise Index on Slice Thickness change with the Rx'd Noise Index. This will assure that when the users changes slice thickness and then returns to the original slice thickness the Noise Index will default to the original value.

- System recommends optimize to 35 mA
- Show Localizer lines are in wrong location if user skips series quickly
- DLP is higher than expected caused by selecting Repeat Series Quickly.
The dose reported was incorrect.

Improvements

- Head recon is slow
- Apostrophe in Hospital Name causes Tube Warm Up to fail
- SmartPrep Diagnostic delay not as expected.

The diagnostic delay is adjusted at Confirm to account for table movement required between the Monitor and Scan phase locations. If the Diagnostic Delay prescribed is less than what the system can do, the expected delay will be posted at Confirm.



- SmartPrep fails to start
- Improvements have been made in communication between the ScanRx and Display Processes to assure reliable use of SmartPrep
- Scan aborts due to Axial not ready
- Start Locations have odd numbers.
Adjustments to Start Locations from Graphic Rx will be rounded off in .25mm increments. This will give more meaningful scan locations for the user.
- Display hung with 10% loading
- System Hung with Can not dismiss Attention Window
- Images can't be retrieved from PACS
- System shuts down with a ScanRx crash
- Dcserver crashes with Storage Commitment requests
- Auto Voice fails to play if the table is moved during a long prep delay
- Images are filmed cut off
This was caused when filming images with GSE from a non square viewport.

- Update Apache HTTP server to be more secure
- Clean up Pro Diags so Reli Monitor does not fail

Enhancements:

SmarTube

There are further enhancements to the SmarTube feature in this release. Additional software has been added to make a selection for tube identification based on tube type,

1. GE-Supplied x-ray tube (Performix)
2. All other tube types.

If a Non-GE-Supplied tube is found, a warning dialog box will be posted at system start-up (see Appendix C).

Advisory messages will be posted on systems with Non-GE-Supplied tubes during Fast Cal and on the Dose Report screen save image indicating that information may not be totally accurate because the full characteristics of the tube may not match those of a GE-Supplied x-ray tube. If the installed x-ray tube is a GE-Supplied tube but the field service engineer did not have all the appropriate information to complete verification, a message (see Appendix C) will be displayed for 30 days or until the field service engineer is able to complete tube installation verification within the 30 day window.

If the verification is not completed in the 30 day window, software will revert to unrecognized tube state and messages for non-GE supplied x-ray tube will be seen during FastCal and on the Dose Report screen save image.

Dose Reporting

Dose Report Screen Save feature is implemented in this release. The Dose Report is a screen save image that is automatically saved to Series 999 at normal End Exam. The Dose Report is a copy of the information displayed in the Dose Information Area on the View Edit screen during the scan. The report includes the CTDIvol, Dose Length Product (DLP) and total DLP for the exam. Dose Report for Scouts will be blank. There is no recognized method for measuring dose for scouts. If there is more than one group in a series, the series will be listed multiple times showing the dose information per group.

For customer inquiries about CTDIvol and DLP, refer the customer to the Dosimetry section in the Technical Reference Manual for their system.

The Dose Report Screen Save can be networked, archived and filmed as part of the patient's record. The Dose Report can be set to automatically auto film on the Film AutoFilm set up page on the View Edit screen. The Dose Report is automatically transferred if Auto Transfer by Exam is selected.

Enhanced Rotor Management:

This feature optimizes rotor usage. The change in rotor management will help prolong X-ray tube life.

This feature is enabled automatically if the system recognizes that a GE supplied X-ray tube is installed. This feature is a purchasable option for tubes not supplied by GE.

Data Logging:

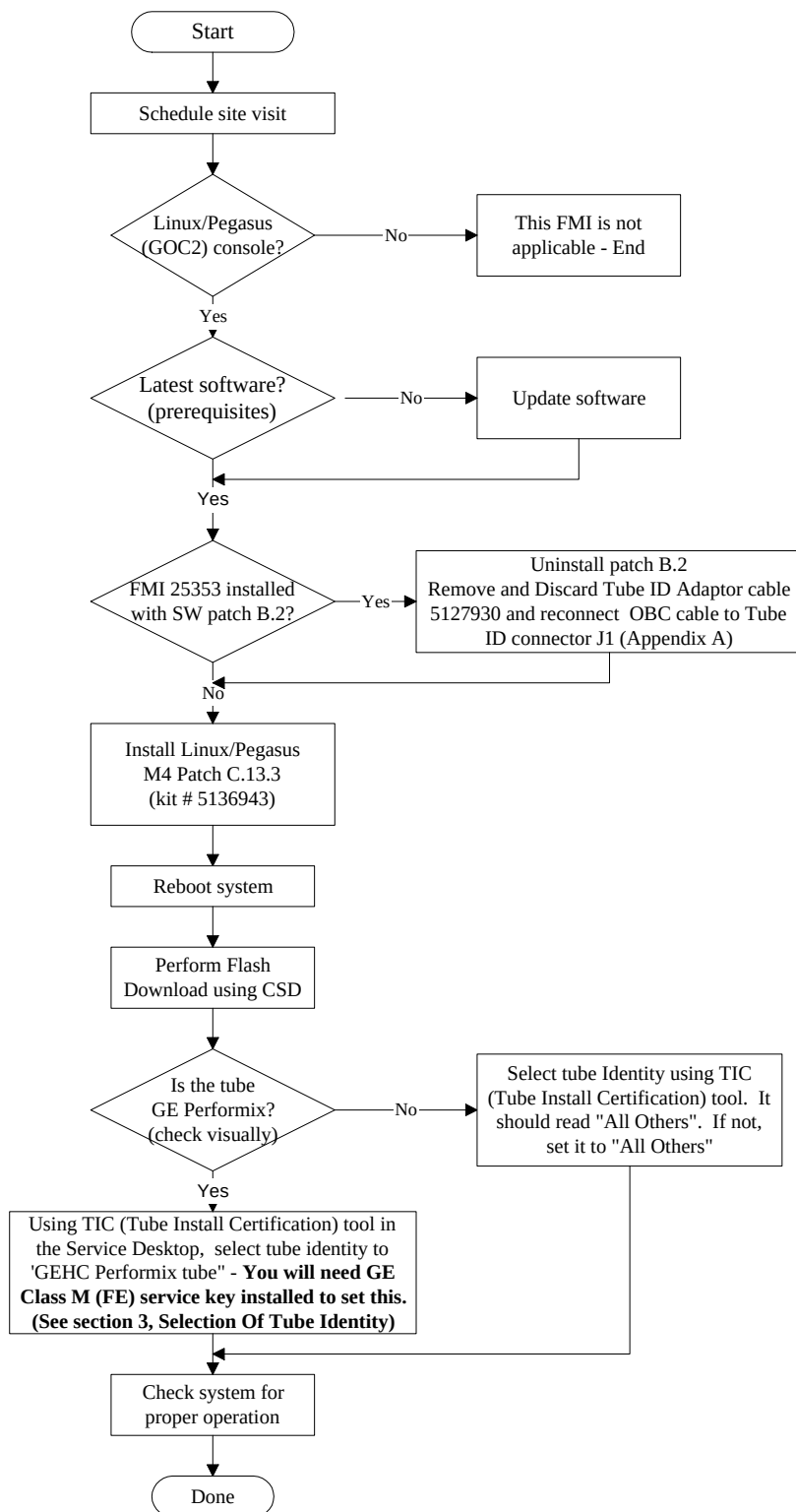
ProDiags Enhancement to include tube usage and RTS. (See section 4 Data Logging Setup)

Product Network Filters (PNF):

Networked medical imaging systems such as CT can be vulnerable to security risks. This software patch includes Product Network Filters (PNF) which addresses the concern of network security by providing a host-based firewall with the capability to filter and block unwanted network communication. The default configuration will be adequate for most all systems. For more detail regarding PNF and configuration information, please refer to Appendix D.

Quality fixes included:

- Reported DLP (dose length product) at the end of the exam is incorrect when returning to view edit to review dose information when AutomA is used.
- Improved transition time from Monitor phase to Scan phase in SmartPrep
- Improvements in Show Localizer Performance
- Auto mA turns off if Next Series is selected too quickly
- Advanced Vessel Analysis has been updated to address measurement inaccuracies

FMI Install Process Flow:

Procedure

1 Preliminary Checks

1. Check to make sure that the system has the latest software. The latest software versions are listed in the Prerequisites section of this document. If the system does not have the latest software, please update the software before installing this FMI

Notice:

- **Install this FMI only in Linux/Pegasus (GOC2) systems.**
- **Do not install this FMI to CT/PET systems**

2. Verify and record patches currently installed.

Open Unix Shell

Type **more /usr/g/patch/inst_patches** <Enter>

 (Record updates info here.)

3. **If the system has been updated with software patch M4 update B.2, Stop here. Execute instructions given in Appendix A to remove patch B.2 before installing patch C.13.3. Any other patches, if exists, need not be removed.**

2 Installation of Software update M4 C.13.3

Installation of patch C: You need the CDROM M4 C.13.3

1. Shut down all CT Applications
2. Open a Unix Shell.
3. Type **su -** <Enter>
4. Type password **#bigguy** <Enter>
5. Insert Linux/Pegasus M4 Patch C.13.3 CD (5154649) into the **Host CT CD drive**.
6. Type **mount /mnt/cdrom** <Enter>
7. Type **cd /mnt/cdrom** <Enter>
8. Type **./install_patch -p grem4.13_patch_C-13-3.i386.rpm** <Enter>

Update software will start loading; information will be displayed regarding the loading.
This process takes ~ 2 minutes.
Text will end with "POST-INSTALL"

9. Type **cd /** <Enter>
10. Type **umount /mnt/cdrom** <Enter>
11. Type **eject** <Enter>
12. Remove the CD from the disk drive
13. Type **reboot** <Enter>
System will go down and then come back to applications.

14. When back to Applications level, open a UNIX shell
15. Confirm that the patch has been successfully installed by typing:

showprods | grep grem4.13_patch_C <Enter>

The output will be similar to:

```
I grem4.13_patch_C 01/12/06 grem4.13_patch_C ver 13 rel 13_3
```

16. Close Unix Shell
17. Open Common Service Desk Top (CSD) and run Flash Download.
 - Select Utilities
 - Select Install Folder
 - Select Flash Down Load Tool
 - Select Update
 - Dismiss when done
18. Select Pink Shutdown button
 - Select Restart
20. When system returns to Applications, select tube identity following step 3.

PLEASE NOTE

- After each re-boot, 'un-recognized X-Ray Tube' message will be displayed as shown in the Appendix C since the tube identity has not been selected. Also displayed on CSD Homepage.
- If there is a need to uninstall software C.13.3, see Appendix B for instructions.

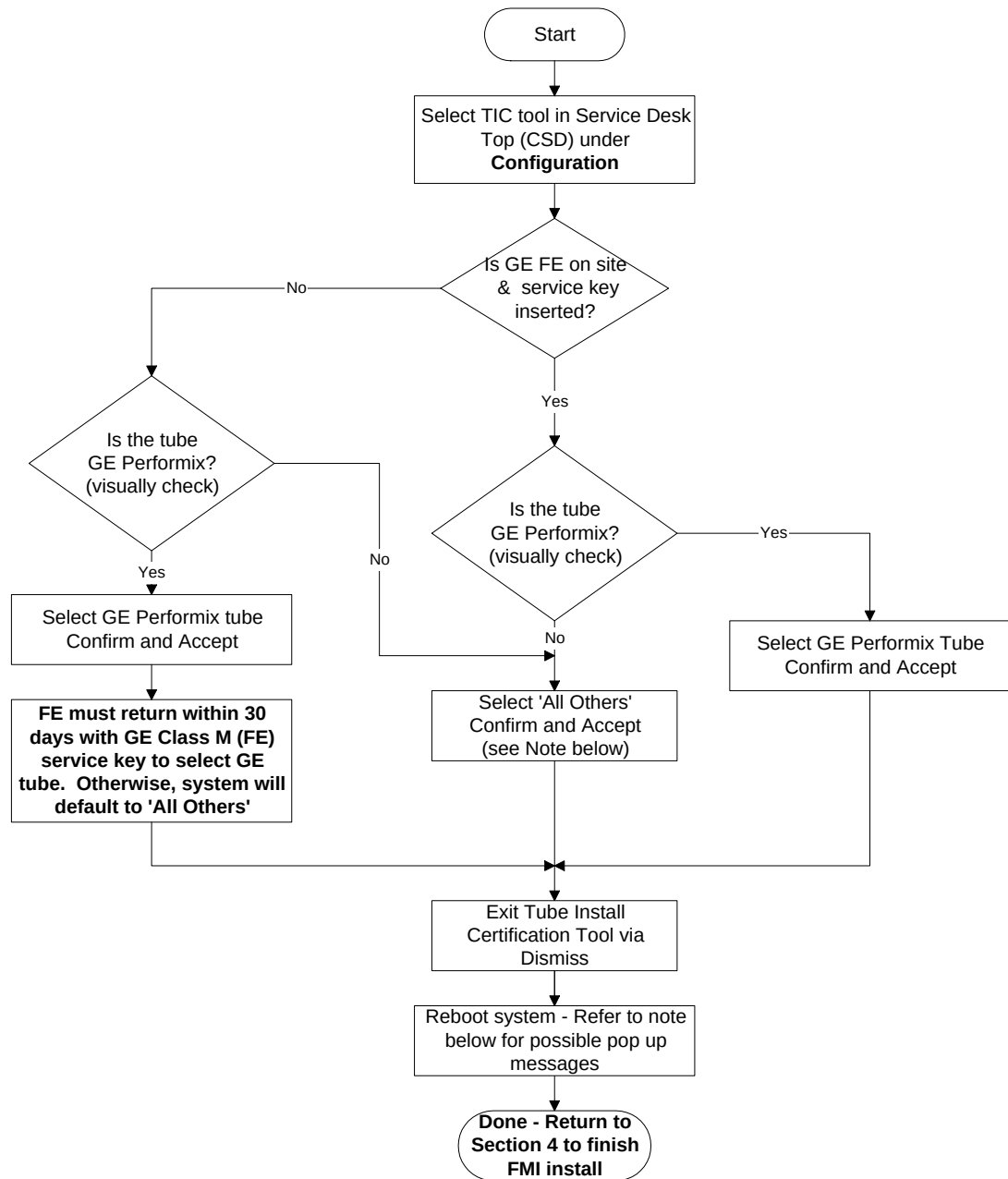
3 Selection of Tube Identity

Select Tube Identity ('GE Medical Systems Performix Tube' or 'All Others') using the Tube Install Certification Tool. The tool is in Common Service Desk Top under **Configuration**. Follow the process shown in the Process Flow on the following page.

Verification of GE Medical System Performix tube: *This can be verified by checking the tube insert model number (2120785 or 2120785-2), which can be found on tube rating plate.*

Note: Do not double click the Tube Certification Tool. This will cause more than one tool window to be displayed, one with message 'service key information not available'. If this happens, close window and try again. If this happens more than once, re-boot system and try again.

Selection of Tube Identity



Note: Possible Pop Up messages at start up:

1. If the tube selection is GE tube without time limit, no pop up messages relating to tube identity on boot up.
2. If the tube selection is GE with time limit, a message with days remaining for GE FE to verify tube will be shown on boot up until GE FE removes the time limit with service key. If not verified by GE FE within the time limit, the system will default to 'All Others'.
3. If the tube is set to 'All Others', un-recognized x-ray tube installed message will always be shown at boot up.

4. Data Logging Setup:

C.13.3 includes data logging enhancements that pushes data to the headquarters for analysis. These tasks are enabled if Class M software is installed and connectivity exists. These new tasks are PushTubeUsage and PushRTS.

The default schedules for these tasks provided in this release are optimized for most of the sites and we recommend use of these default schedules. Careful planning is required before making any schedule changes as the amount of data pushed can vary based on the frequency. Changing the schedule may result in system slowing down dramatically due to excessive amounts of data pushed at once, or data overwritten if not pushed frequently enough.

To check ProDiags, open unix shell and type: **prodiags** <Enter>

If you get Netscape warning messages, ignore. No impact to the operation of prodiags.

ProDiags Screen:

Proactive Diagnostics

Schedule Task Time Slot Log & Result

Schedule Management

Schedule Management provides the following functionalities

1. Add a task to the schedule
2. Delete a task from the schedule
3. View the schedule

Task	Iterations	Day	Hour	Minute	Type	Time Slot
PassChg	1	Daily	05 AM	30	Background	
pd_HouseK...	1	Daily	04 AM	30	Background	
pd_ASC_No...	1	Daily	Hourly	28	Background	
healthpg	1	Daily	04 AM	45	Background	
IIP_HouseK...	1	Daily	06 AM	12	Background	
IIP_HouseK...	1	Daily	12 PM	12	Background	
IIP_HouseK...	1	Daily	06 PM	12	Background	
IIP_HouseK...	1	Daily	12 AM	12	Background	
idm_send...	1	Daily	07 AM	00	Background	
idm	1	Daily	Hourly	00	Background	
ctrdi	1	Daily	12 PM	59	Background	
PushTubeU...	1	Wednesday	10 AM	16	Background	
PushRTS	1	Daily	10 AM	07	Background	

New Tasks

New Delete Reload Apply Help

Status: Reading data... Done

5 System Checkout

Perform the following scans (Abdominal or Head)

50 images at 0.5 sec rotation at 120KV, 40mA

The system should scan and image without any errors

6 Finishing Up

1. Make sure that the 'Tips and Workaround' document, 5130011-100 and the Customer Letter, 5142333-147 (the correct language version) supplied with this FMI kit are left with the customer.
2. Close FMI per regional procedure. Note that the FMI cannot be closed until the Product Locator cards are completed, sent in, and GIB is updated with all tracked items in this FMI.
3. Remove this FMI document from the customer site.

Appendix A

Removal Of B.2

B.2 Uninstallation:

1. Shutdown all CT applications
2. Insert Linux Pegasus M4 B.2 CD into the CD drive
3. Open a unix shell
4. Type **su** - <Enter>
5. Type password **#bigguy**
6. Type **mount /mnt/cdrom** <Enter>
7. Type **cd /mnt/cdrom** <Enter>
8. Type **./uninstall_patch** <Enter>
- 9.

This will display a menu similar to the one shown below with all the available patches installed.

Screen similar to below will be shown.

Select 0 to exit.

0 - exit

1 - Lnx_Peg_M4__B_2

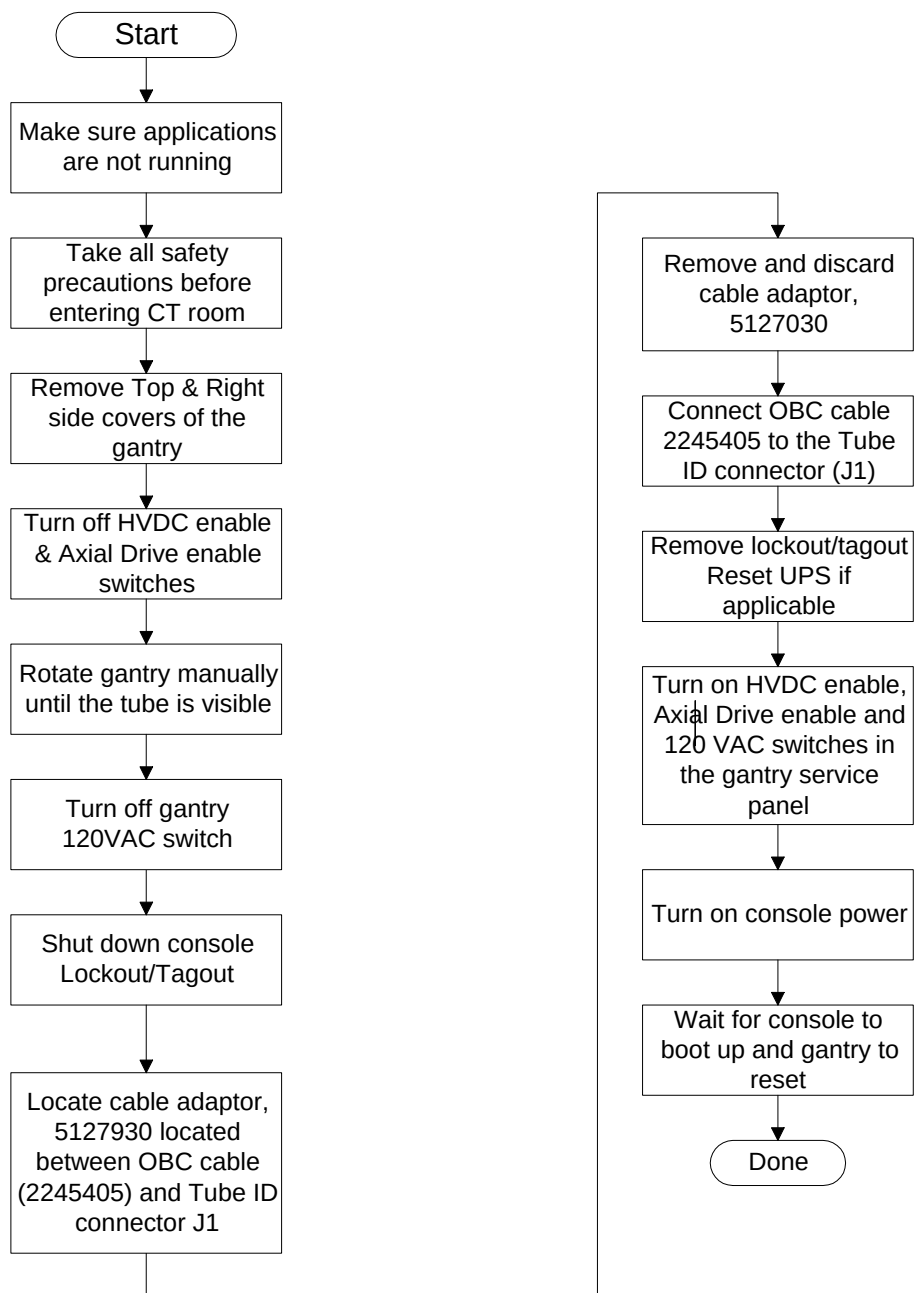
Enter a number:

Please select the number associated with the patch to be uninstalled.

10. Type **(select appropriate number for B.2 from above)** <Enter>
11. Type **cd /** <Enter>
12. Type **umount /mnt/cdrom** <Enter>
13. Type **eject** <Enter>
14. Remove the CD out of the disk drive.
15. Type **reboot** <Enter>
16. Once the system has rebooted run Flash Download (see step 14 of the install Instructions)
17. Select Pink Shutdown button
Select Restart

Appendix A – Contd.

Removal of Cable Adaptor, 5127930



Appendix A – Contd.

Removal of Cable Adaptor, 5127930

1. Make sure that applications are not running
2. Take all safety precautions before entering the scan room
3. Lower table to home (lowest) position
4. Remove gantry top and side covers

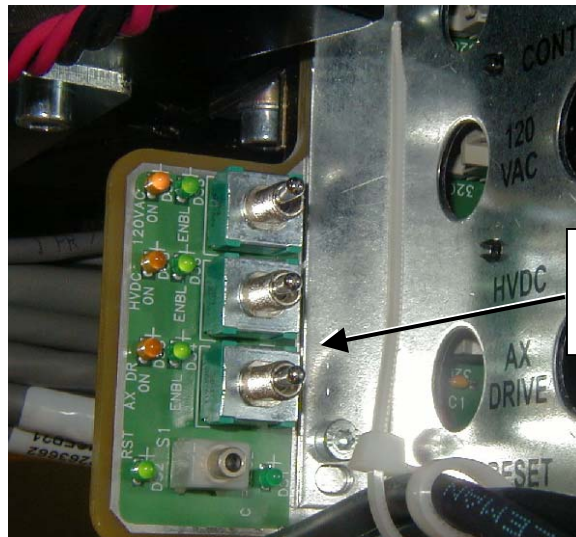


WARNING



FAILURE TO DISABLE THE AXIAL DRIVE MAY RESULT IN SERIOUS INJURY DUE TO UNEXPECTED GANTRY ROTATION.

5. Turn off high voltage DC enable switch (HVDC) and the Axial Drive enable (AX DR) switch to off position in the service panel inside the gantry

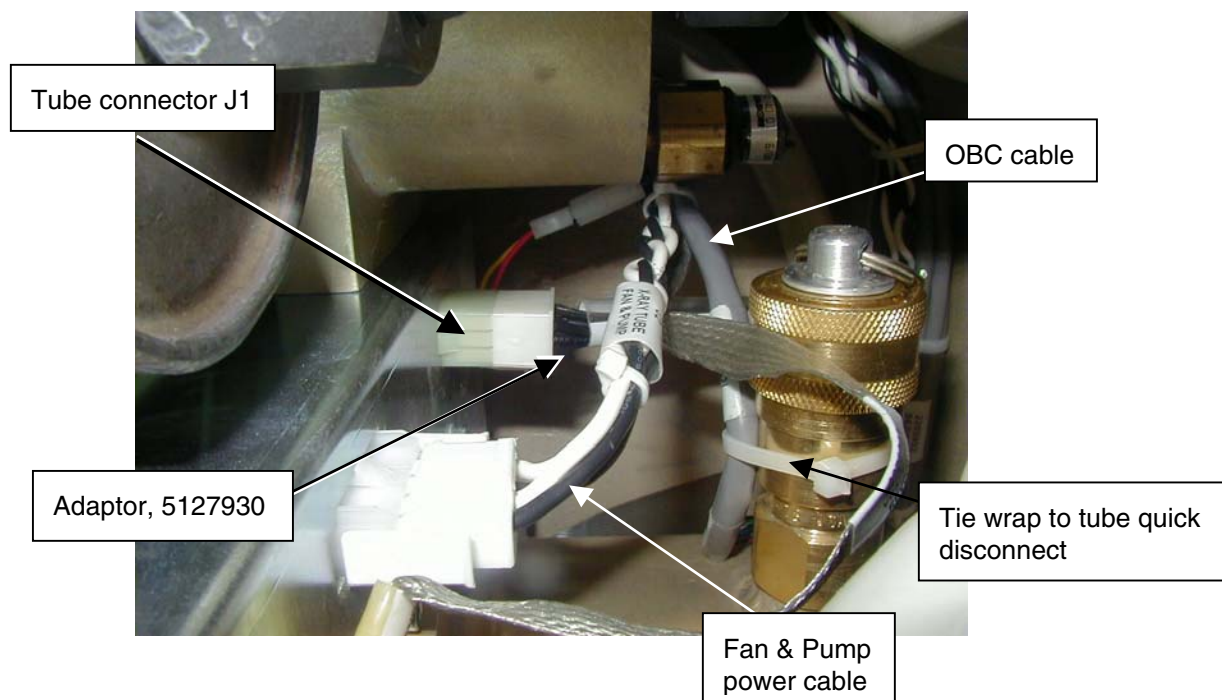


HVDC enable,
Axial Drive enable and
120VAC switches

6. Rotate the gantry with care until the tube is visible.
7. Now, turn off the 120VAC switch in the service panel
8. Shut off power to the console and follow lockout/tag out procedure to make sure that power is off while making the changes to the gantry
9. Locate cable adaptor, 5127930 which is connected to OBC cable, 22445405 at one end and to the Tube ID connector J1 at the other end. It is behind the tube cooling tank power cable.
10. Remove the cable adaptor, 5127930 and discard.
11. Connect cable 2245405 to the tube at connector J1.
12. For easy access to tube connector J1, if you have to disconnect Fan & Pump power cable, **REMEMBER TO RECONNECT AT COMPLETION.**

Appendix A – Contd.

Removal of Cable Adaptor, 5127930



IMPORTANT: Visually check the gantry to make sure that:

- ***The power cable to the tube cooling tank is connected***
- ***The tube ground wire is in place***

13. Remove lockout/tag out so that power can be applied to the system. If the site uses UPS, reset UPS.
14. Turn on all three switches (Axial Drive, HVDC and 120VAC) in the gantry
15. Power up console
16. Wait for the console to boot up and the gantry to reset

Return to section 2, Installation of Software M4 B.X

Appendix B

Removal Of C.13.3.

Note:

If you uninstall C.13.3, you need to restore system to the original system configuration. Patches which were installed before C.13.3 should be restored.

C.13.3 Uninstallation:

1. Shutdown all CT applications
2. Insert Linux Pegasus M4 C.13.3 CD in the drive
3. Open a unix shell
4. Type **su** - <Enter>
5. Type password **#bigguy**
6. Type **mount /mnt/cdrom** <Enter>
7. Type **cd /mnt/cdrom** <Enter>
8. Type **./uninstall_patch** <Enter>

This will display a menu similar to the one shown below with all the available updates installed.

The installed patches are listed below.

Select 0 to exit.

0 - exit

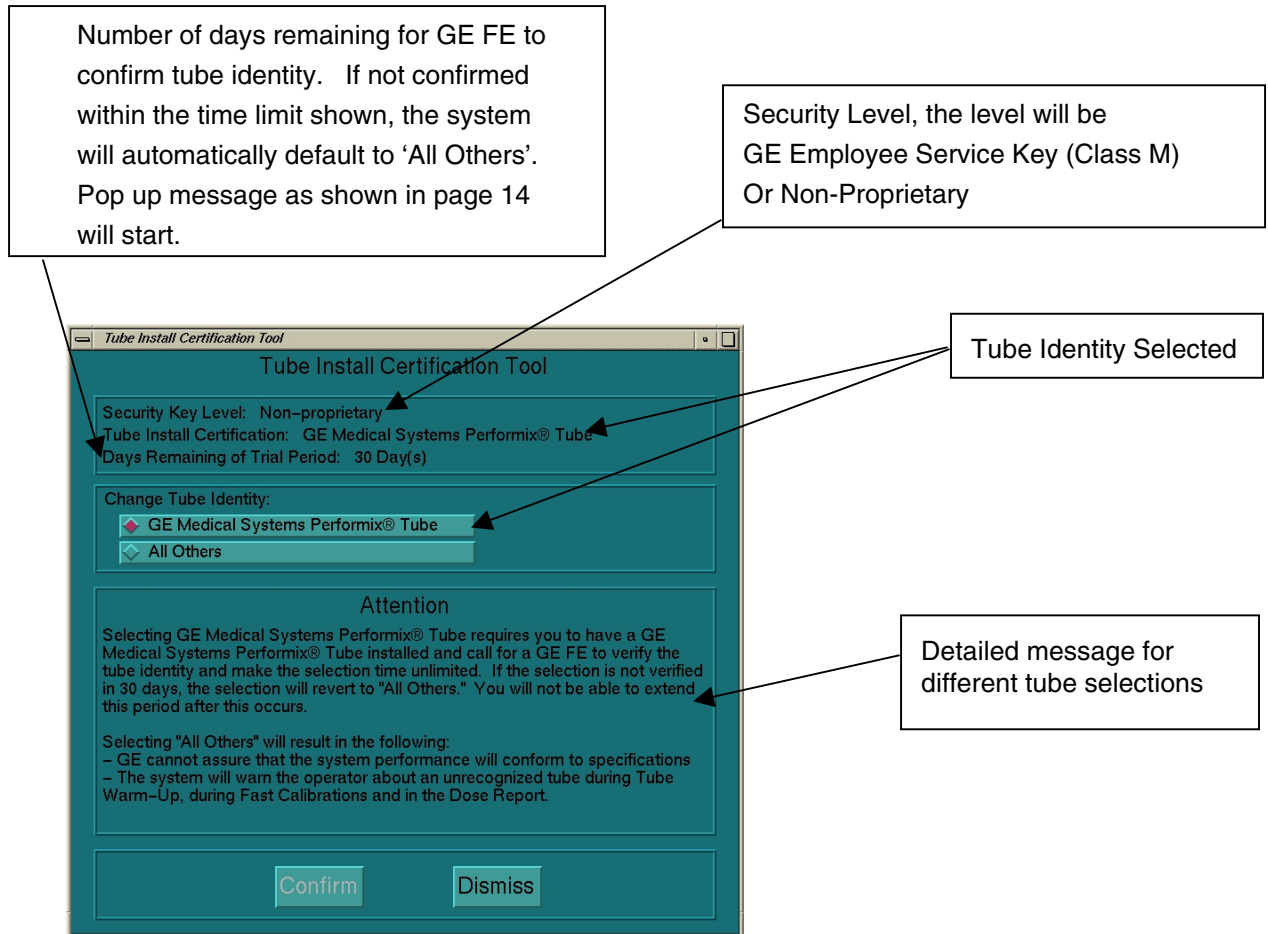
1 - grem4.13__ C-13-3

Enter a number:

Please select the number associated with the patch to be uninstalled.

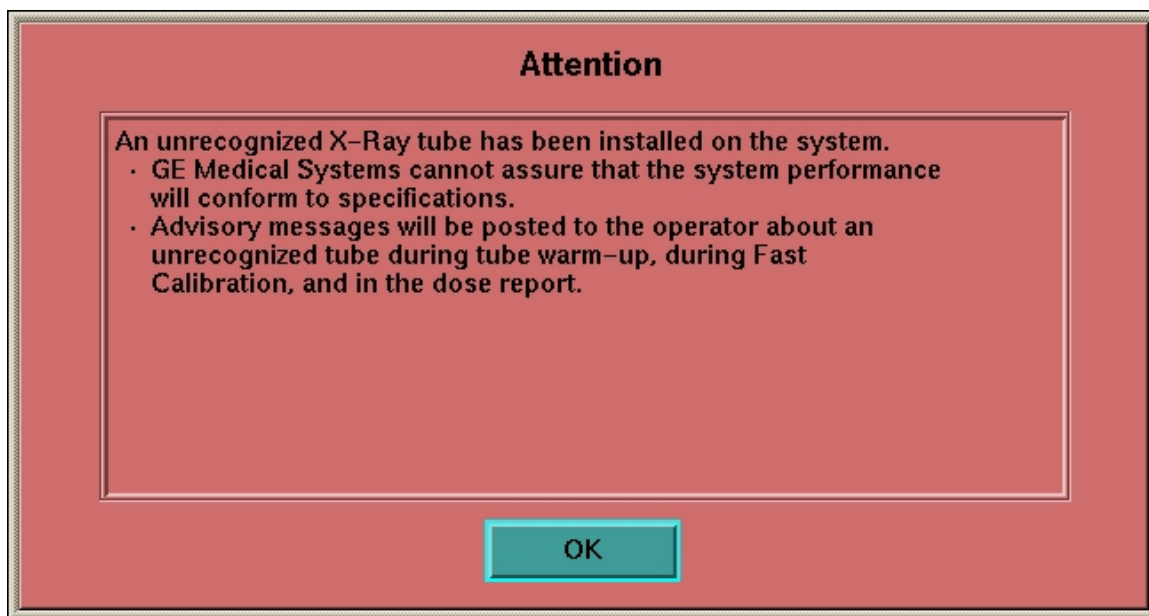
9. Type **(select appropriate number for C.13.3 from above)** <Enter>
10. Type **cd /** <Enter>
11. Type **umount /mnt/cdrom** <Enter>
12. Type **eject** <Enter>
13. Remove the CD out of the disk drive.
14. Type **reboot** <Enter>
15. Once the system has rebooted run Flash Download (see step 14 of the install Instructions)
16. Select Pink Shutdown button
Select Restart

Appendix C - Screen Displays



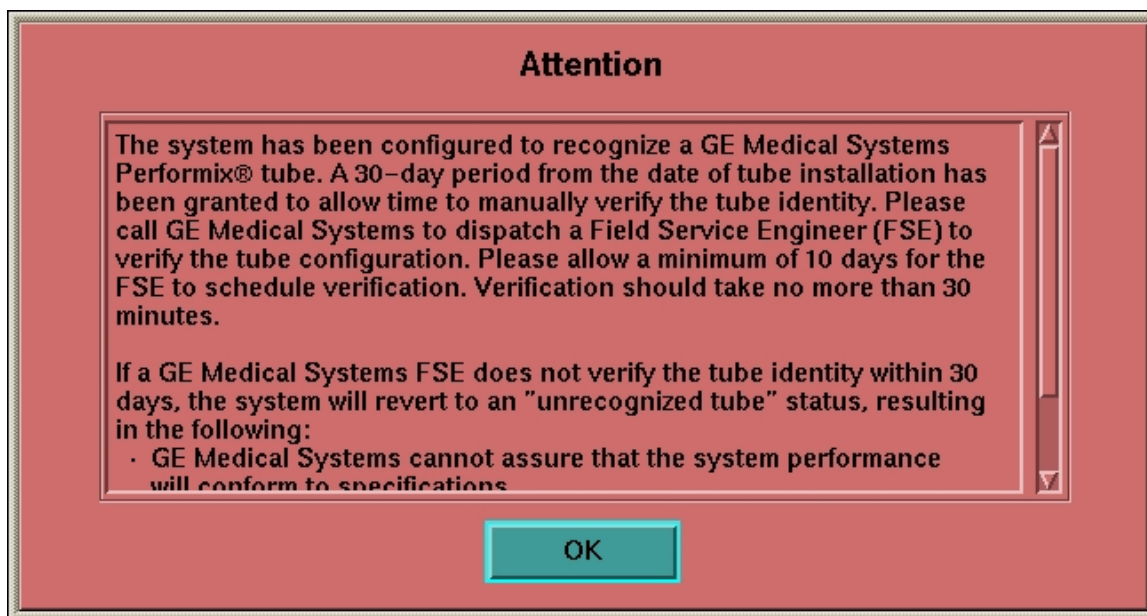
Example of Tube Install Certification Tool

Appendix C - Screen Displays (Contd.)



Message at each boot up when unrecognized X-Ray Tube is installed in System

Appendix C - Screen Displays (Contd.)



**Message at each boot up when GE Tube is selected but not verified by
GE Field Engineer**

Appendix C - Screen Displays (Contd.)

Patient Name: Sam Ample				Exam no: 507	
Accession Number:				21 Jun 2005	
Patient ID: Sample				LightSpeed Plus	
Exam Description:					
Dose Report					
Series	Type	Scan Range (mm)	CTDIvol (mGy)	DLP (mGy-cm)	Phantom cm
1	Scout	-	-	-	-
2	Helical	S30.000-S7.500	29.75	106.93	Body 32
2	Helical	I72.500-I95.000	29.75	106.93	Body 32
Total Exam DLP:				213.86	
1/1					
W:1 L:-1025					

Dose Report Screen when GE Medical Systems
X-Ray Tube is installed in System

Appendix C - Screen Displays (Contd.)

Patient Name: Test Name				Exam no: 504	
Accession Number:				21 Jun 2005	
Patient ID: sfsf				LightSpeed Plus	
Exam Description:					
Dose Report					
Series	Type	Scan Range (mm)	CTDIvol (mGy)	DLP (mGy-cm)	Phantom cm
1	Scout	-	-	-	-
2	Helical	S30.000-S7.500	29.75	106.93	Body 32
Total Exam DLP:				106.93	
Unrecognized tube in use - Dose may vary. The reported dose information is calculated based on empirical observations of systems with GE Medical Systems Performix® tubes. GE cannot assure the accuracy of reported dose information for any configurations that include tubes other than GE Medical Systems Performix tubes.					
1/1					
W:1 L:-1025					

Dose Report Screen when un-recognized
X-Ray Tube is installed in System

Appendix D

Product Network Filter (PNF)

Networked medical imaging systems such as CT can be vulnerable to security risks. This vulnerability can also inhibit remote service connectivity. Product Network Filters (PNF) is a software firewall tool that allows a site's Network Administrator to control traffic from their internal network to the diagnostic imaging system.

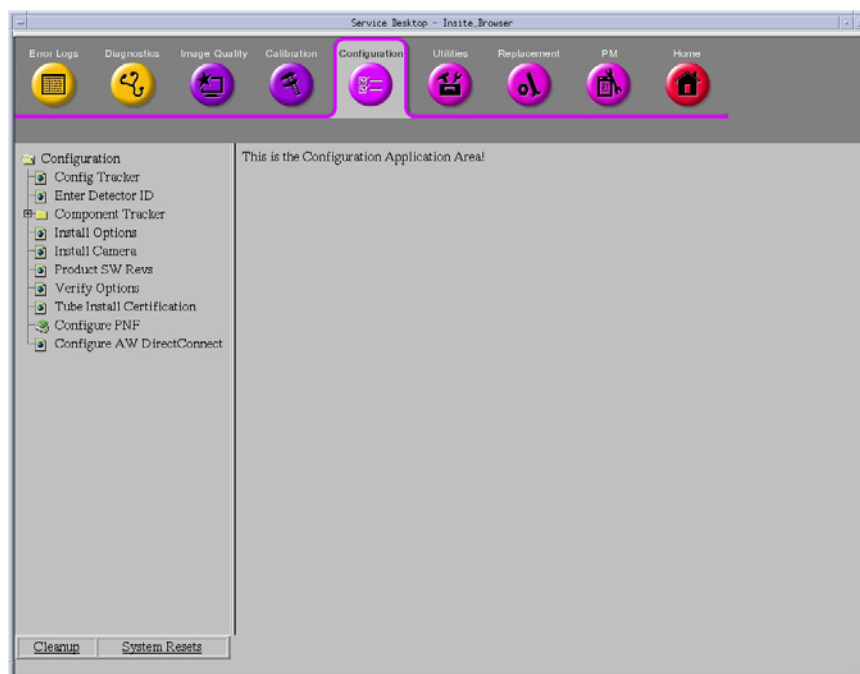
PNF addresses the concern of network security of imaging systems by providing a host-based firewall that filters and blocks network communication. Anyone who tries to access CT from a particular network address will need to go through PNF. PNF "listens" to network traffic and decides if that network address should be able to access CT or not.

PNF was created to:

- Respond to customer concerns regarding Diagnostic Imaging (DI) product security
- Secure the DI product network interface (including remote service)
- Significantly reduce the risk of virus/worm attack on the DI products

The default values for PNF should be adequate for all systems. Should you need to check configuration,

Product Network Filter configuration can be accessed via the Common Service Desktop (CSD) using the following screen. Select Configuration, the Configure PNF.



Note: Firewall should be kept on at all times.

However, if you feel you have a network problem related to PNF, you can disable PNF, retry your network. If network problem still exists, **return to PNF and re-enable Firewall!**

If problem disappears when firewall is turned off. Contact OLC or Connectivity Team for help in configuration.

Product Network Filters - Insite Browser

Product Network Filters

Firewall: ☒ On ☐ Off

PNF Version: 2.3.5

Named Services | Allowed Nodes | DICOM | Expert | GE Service | Debug | Help

Named Services

	Service	Allowed IP's
☐	ping (icmp)	*
☐	telnet	*
☐	telnet	3.231.48.9
New	- Services	

Firewall Settings